Question	Answer	Marks	Partial Marks
5(b)(i)	$\frac{95}{y}+\frac{147}{y+2}=12$	**M2**	**M1** for $\frac{95}{y}$ or $\frac{147}{y+2}$
	$95(y+2)+147y=12y(y+2)$ oe	**M1**	Allow correct or for clearing *their* equation with algebraic fractions in y and $y + 2$ Allow $95y+190+147y=12y^2+24y$ oe
	leading to $6y^2 – 109y – 95 = 0$	**A1**	With all brackets shown expanded and no errors or omissions
5(b)(ii)	$(6y + 5)(y – 19)$	**2**	**B1** for $(6y + a)(y + b)$ with $ab = –95$ or $a + 6b = –109$ or $(3y + a)(2y + b)$ with $ab = –95$ or $2a + 3b = –109$ or for partial factorisation $y(6y + 5) – 19(6y + 5)$ or $6y(y – 19) + 5(y – 19)$
5(b)(iii)	19	**1**	Correct or FT *their* positive answer from factors dep on B1 earned
6	11.9 or 11.91 to 11.92	**7**	**B5** for $t = 1.055$ or 1.0550... **M1** for $\tan w=\frac{their\ t}{5}$ oe OR **M1** for $(2t+3)^2=t^2+5^2$ oe seen isw **M2** for $3t^2+12t-16[=0]$ oe seen isw or **B1** for $4t^2+6t+6t+9$ **M1FT** for $\frac{-12\pm\sqrt{12^2-4(3)(-16)}}{2(3)}$ oe **M1** for $\tan w=\frac{their\ t}{5}$ oe
7(a)(i)	21.5 or 21.52...	**2**	**M1** for $\tan(\ldots)=\frac{2.8}{7.1}$ oe
7(a)(ii)	10.2 or 10.17 to 10.18	**3**	**M2** for $\left(\frac{2.8}{\tan 21}\right)^2+7.1^2$ oe or **M1** for $\frac{2.8}{PR}=\tan 21$ oe

Question	Answer	Marks	Partial Marks
7(b)	76.5 or 76.52 to 76.53	**3**	**M2** for $[\sin =]\frac{16.7\sin 32}{9.1}$ oe or **M1** for $\frac{9.1}{\sin 32}=\frac{16.7}{\sin M}$ oe
7(c)(i)	$\frac{1}{2}\times 12.3\times 21.5\sin(...)=62.89$ or better	**M1**	
	28.40 to 28.41…	**A1**	
7(c)(ii)	12.2 or 12.17 to 12.18	**3**	**M2** for $\sqrt{12.3^2+21.5^2-2\times 12.3\times 21.5\times\cos 28.4}$ OR **M1** for $12.3^2 + 21.5^2 – 2 \times 12.3 \times 21.5 \times \cos28.4$ **A1** for 148 or 148.2 to 148.3
7(c)(iii)	6.6[0] to 6.62	**3**	**M2** for 21.5cos28.4 – 12.3 or **M1** for 21.5cos28.4
8(a)(i)	$\frac{1}{6}$ oe	**1**	
8(a)(ii)	25	**1**	**FT** *their* **(a)(i)** dep on 0 < **(a)** < 1
8(b)(i)	$\frac{11}{36}$ oe	**3**	**M2** for $\frac{1}{6}\times\frac{2}{6}+\frac{3}{6}\times\frac{3}{6}$ oe or correct possibility diagram with 11 outcomes identified or **M1** for $\frac{1}{6}\times\frac{2}{6}$ or $\frac{3}{6}\times\frac{3}{6}$ oe or lists the 11 required outcomes or for possibility diagram but required outcomes not indicated
8(b)(ii)	$\frac{2}{11}$ oe	**2**	**M1** for $\frac{2}{k}$ or $\frac{p}{their\,11}$ seen oe leading to answer
8(c)	6	**2**	**M1** for $\left(\frac{4}{6}\right)^k\times\frac{2}{6}=\frac{32}{729}$ written oe soi by one trial with $k > 1$ or $2^{n-1}=32$ or better or $3^n=729$ or better
9(a)	$12x^2-4x^3$ oe final answer	**2**	**B1** for $12x^2$ or $-4x^3$ in final answer or for correct answer seen